MASKER : Masked Keyword Regularization for Reliable Text Classification

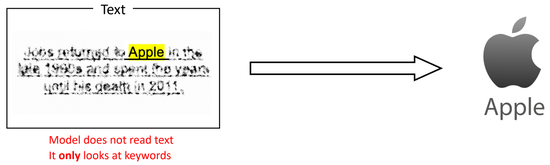
SeungJun Moon*[1,2], Sangwoo Mo*[1], Kimin Lee[3], Jaeho Lee[1], Jinwoo Shin[1](* equal contribution)

[1]KAIST, [2]Klleon Inc., [3]UC Berkeley

TL; DR. We propose a simple yet effective regularization method for language models which enhances out-of-distribution(OOD) detection and cross-domain generalization performance

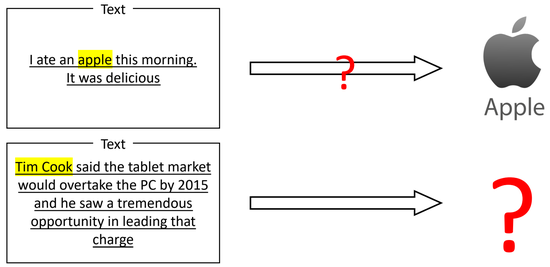
Introduction

Language models can find easy ways for classification by only using **keywords**, which does not harm classification accuracy



However, for OOD detection and cross-domain generalization settings, over-reliance on keywords is problematic:

- OOD data might contain keywords, which might lead to over-confidence
- Cross-domain data might contain keywords not appeared in the training set

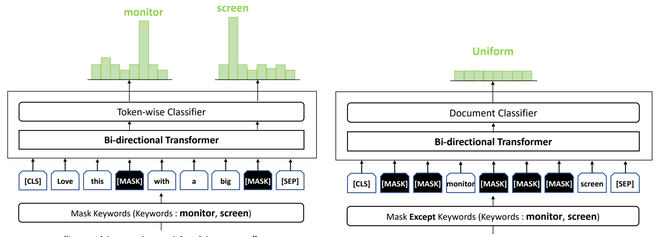


Contribution

- We propose a regularization scheme **MASKER**, which consists of (a) **Masked Keyword Reconstruction(MKR)** and (b) **Masked Entropy Regularization(MER)**
- We suggest keyword selection schemes which is used for MKR and MER
- We demonstrate that MASKER applied to BERT, RoBERTa, and ALBERT significantly improves OOD detection and cross-domain generalization performance

MASKER

MASKER consists of **MKR**(left) and **MER**(right)



MKR aims to reconstruct masked keywords. MKR is done by the following process:

- Mask keywords in the input text with a probability 0.5
- Force the model to guess the keywords by using context around it

MER aims to generate OOD data containing keywords and forces model to classify it as an OOD. MER is done by the following process:

- Mask every token except keywords
- Regularize document classifier to yield uniform distribution for inputs

Keyword Selection

We select keywords with two methods:

- **Frequency-based method** uses score s^{freq} for token t:

$$s^{\texttt{freq}}(t) = \max_{c \in \{1,\dots,C\}} \texttt{tf}(t, \mathbf{X}_c) \cdot \texttt{idf}(t, \mathbf{D})$$

where $\texttt{tf}(t, \mathbf{X}) = 0.5 + 0.5 \cdot n_{t,\mathbf{X}} / \max\{n_{t',\mathbf{X}} : t' \in \mathbf{X}\}$

$\texttt{idf}(t, \mathbf{D}) = \log(|\mathbf{D}|/|\{\mathbf{X} \in \mathbf{D} : t \in \mathbf{X}\}|)$

- **Attention-base method** uses score s^{attn} for token t:

$$s^{\texttt{attn}}(t) = \sum_{(\mathbf{x},y)\in\mathcal{D}} \frac{1}{n_{t,\mathbf{x}}} \sum_{i \in \{1,\dots,T\}} \mathbb{I}(t_i = t) \cdot \frac{a_i}{\|\mathbf{a}\|}$$

where $\boldsymbol{a} = [a_1, \dots, a_T]$ be the attention value, where a_i corresponds to t_i

Main Results

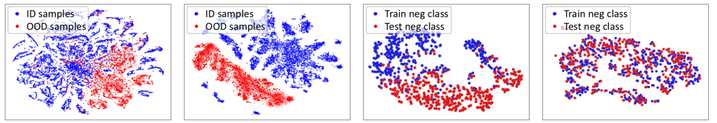
MASKER shows consistent improvement on various OOD detection and cross-domain generalization settings

ID	OOD	OC-SVM	DOC	Vanilla / Residual / MASKER BERT	RoBERTa	ALBERT
Newsgroup	Amazon	62.1	84.1	85.4/86.7/**87.0**(+1.9%)	85.3/85.9/**87.2**(+2.3%)	86.7/85.4/**89.4**(+3.1%)
	Reuters	53.9	60.0	91.8/93.0/**97.7**(+6.4%)	93.1/92.1/**93.9**(+0.8%)	93.3/92.0/**94.7**(+1.6%)
	IMDB	59.8	88.6	94.6/95.7/**98.5**(+4.1%)	95.2/95.9/**97.7**(+2.7%)	94.5/92.6/**96.6**(+2.2%)
	SST-2	63.0	88.1	87.0/97.0/**98.6**(+13.3%)	94.7/94.9/**98.2**(+3.6%)	95.8/95.6/**98.1**(+2.4%)
	Fine Food	62.8	81.3	85.3/87.3/**93.4**(+9.5%)	88.7/89.4/**92.9**(+4.7%)	77.6/86.6/**91.6**(+18.1%)
Amazon	Newsgroup	61.3	81.3	84.8/83.9/**87.2**(+2.8%)	87.9/86.6/**91.0**(+3.5%)	87.3/85.0/**88.4**(+1.3%)
	Reuters	55.5	79.8	89.7/89.7/**93.5**(+4.2%)	92.3/92.6/**93.6**(+1.4%)	93.1/93.5/**94.5**(+1.5%)
	IMDB	66.2	89.6	93.3/92.8/**95.2**(+2.0%)	90.1/87.0/**93.3**(+3.5%)	89.9/88.6/**95.6**(+6.4%)
	SST-2	60.9	91.5	93.0/88.9/**95.6**(+2.8%)	92.4/94.8/**96.4**(+4.4%)	93.4/91.9/**96.9**(+3.7%)
	Fine Food	51.1	66.8	78.5/77.7/**84.9**(+8.2%)	74.9/80.0/**80.7**(+7.8%)	82.6/86.3/**87.3**(+5.7%)

Table 1 : Result on OOD detection

Train	Test	OpenMax	DOC	Vanilla / Residual / MASKER BERT	RoBERTa	ALBERT
IMDB	IMDB	87.7	88.0	93.5/92.8/93.5	95.3/94.0/95.6	91.6/91.4/90.8
	SST-2	79.6	77.9	85.9/86.9/**88.1**(+2.6%)	89.7/90.2/**91.8**(+2.3%)	89.8/89.0/**89.9**(+0.1%)
	Fine Food	75.4	78.3	92.9/92.5/**93.6**(+0.8%)	92.6/87.7/**93.0**(+0.4%)	87.1/87.8/**92.1**(+5.7%)
SST-2	SST-2	82.9	83.1	92.5/90.3/92.3	94.5/92.0/94.3	91.9/90.9/91.5
	IMDB	75.3	76.9	85.7/86.2/**88.4**(+3.2%)	86.0/85.7/**87.3**(+1.5%)	83.0/83.0/**83.8**(+1.0%)
	Fine Food	62.2	64.5	87.6/88.2/**89.2**(+1.8%)	86.5/87.8/**89.6**(+3.6%)	79.0/80.6/**84.9**(+7.5%)
Fine Food	Fine Food	93.6	93.3	96.5/94.8/96.5	96.9/95.7/97.1	95.5/95.4/96.6
	IMDB	67.5	67.3	77.3/81.1/**85.6**(+10.7%)	84.1/84.6/**86.6**(+2.9%)	74.7/80.4/**84.8**(+13.5%)
	SST-2	61.9	62.0	67.6/67.8/**80.0**(+18.3%)	78.5/80.1/**83.8**(+6.8%)	71.7/73.2/**83.3**(+16.2%)

Table 2 : Result on cross-domain generalization



(a) BERT (b) MASKER (ours) (c) BERT (d) MASKER (ours)

Table 3 : t-SNE plot on OOD detection and cross-domain settings

We can see that MASKER distinguishes OOD data, entangles data from the same classes in the cross-domain dataset, compared to vanilla models

Future Work : Can we apply MASKER to language generation models?